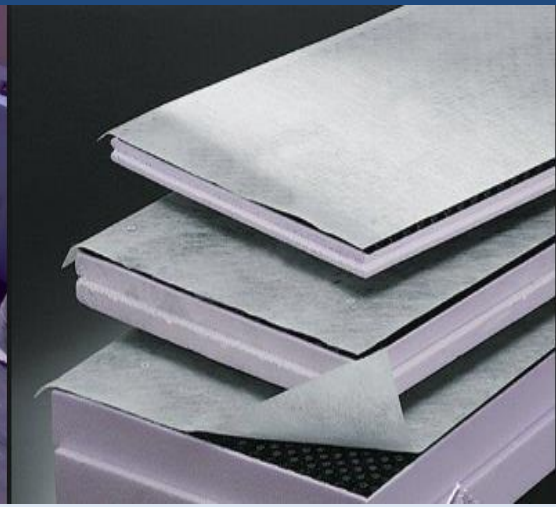


A Pre-feasibility for the Establishment of Polystyrene Panels Manufacturing Plant



2017

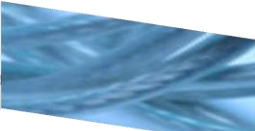


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1. Executive Summary

The project aims to establish a factory for the production of polystyrene panels in accordance with international specifications (pressure 32-35). The proposed location of the project is Irbid governorate and specifically in the area of Al-Hassan Industrial City, which represents the place and environment of the industrial project.

There are factories in Jordan for the production of insulation panels, but the domestic market consumes panels that are imported because of non-compliance to international specifications in the local production, hence the idea of the project is to provide a high return on investment, and will meet the needs and requirements of the local market.

The project, if implemented, will limit the import of these products and achieve self-sufficiency with high quality products and competitive prices in the region.

The following table shows the economic indicators of the project:

Table 1 : Project Main Indicators

Total Investment cost (JD)	2,172,295
Internal Return Rate	%19.71
Payback Period (Years)	7
Net Present Value (JD)	1,485,185
Job Vacancies	21
Annual Revenue Expectations for 10	14,126,400
Total Operating Cost for 10 years (JD)	8,647,563
Gross Profit for 10 years (JD)	1,632,754

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

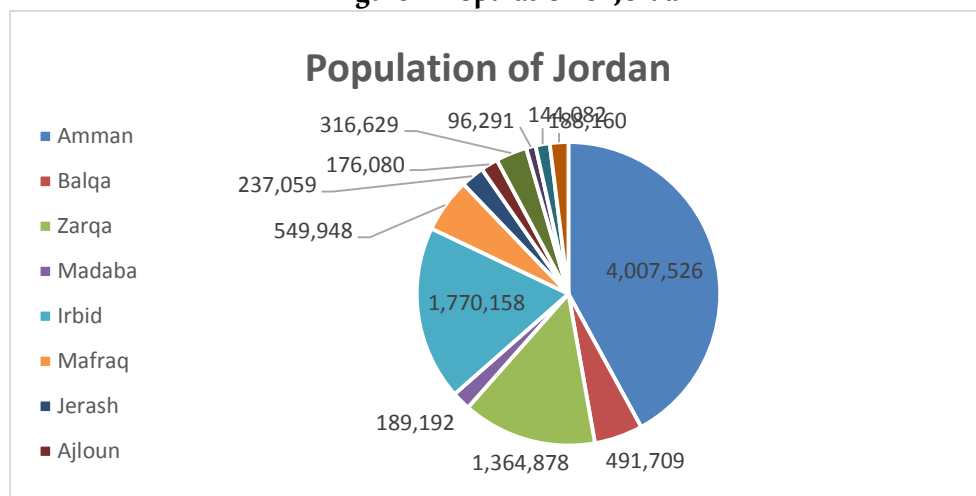
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 : Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

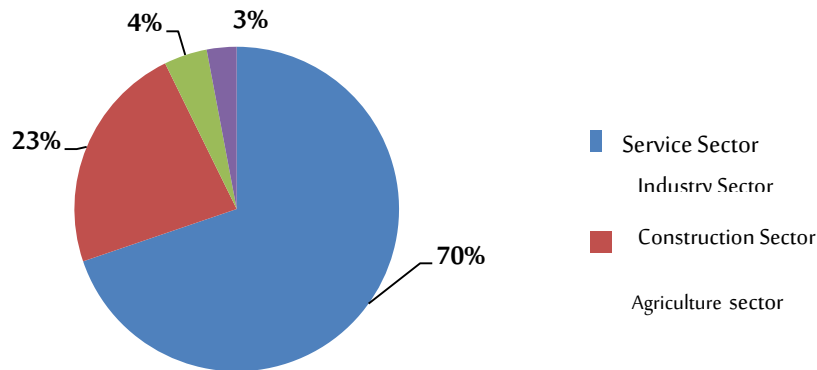
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Irbid:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table3 : Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20% As per Investment Law categorization, Irbid is classified in Class (C) where investors enjoys 60% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.

3. Market Analysis

Project Description

The project aims to establish a plant for polystyrene panels used in thermal insulation in the governorate of Irbid.

These panels are used in buildings to reduce thermal leakage from outside the building to the inside and vice versa, in addition to being an acoustic insulation by shock absorption.

The project was adopted because of the increase in urban activity in the governorate in line with the population increase, in addition to the large demand for insulation materials in order to reduce energy consumption, and since most of the polystyrene used in the domestic market were imported, thus the need to manufacture polystyrene locally has emerged.

Project Objectives

The Objectives of the project are summarized as following:

- Achieve earnings and financial returns to the investors.
- Vitalizing the economic movement in Irbid.
- Exploitation of the market of Irbid due to the increase in population growth and urbanization.
- Reduce energy consumption by either cooling in summer or heating in winter.

The Proposed Service

The proposed product will be the blue extruded polystyrene panels (Pressure 34) used for indoor thermal and sound insulation.

Target Segments

The target segments for the proposed project are the planned housing and buildings of the Kingdom.

Market Size Analysis

The following table shows the number of houses and commercial buildings until 2015 detailed as follows:

Table 4 Housing and buildings Number

Buildings Type	Number
Villa	17094
Buildig	288491
House	489364
Institutions	45927
Hotels	624
Others	239
Total	841,739

Department of Statistics 2015 - General Census of Population and Housing

The following table shows the licenses granted and the area (in 1,000 sqm) for the years (2011-2016):

Table 5 : The Granted Licenses and Areas

	2011	2012	2013	2014	2015	2016
Housing License	26,603	29,940	33,906	36,642	32,882	29,507.8
Area	9,719.9	10,691.2	11,817.6	12,578.4	10,870	9,753.7
Other Purposes	2,973	2526	2695	2936	2893	2,595.9
Licenses						
Area	2,440.7	2,215.5	2,167.5	2,413.9	2,243.3	2,013
Total Number of	29,576	32,466	36,601	39,578	35,775	32,102
Licenses						
Total Area (000 sqm)	12,160.6	12,906.7	13,985.1	14,922.3	13,122.8	11,775

Source: Central Bank - Statistical Tables

The number of licenses and area for 2016 is based on the decline rate between (2014-2015).

- Based on the statistics and studies, the total area of the existing buildings in 2016 amounted to 11,755,000 sqm.
- According to the studies and the market analysis (contractors), 10% of the houses and buildings are using insulation yearly. Accordingly, the area used for insulation in 2016 is 1,176,670 sqm multiplied by the number of facades of Building (4) equals 4,706,680 sqm. Since the facades have windows and doors, the area was calculated by 75% to estimate the final area used for insulating polystyrene which amounted to 3,530,010 sqm.
- One cubic meter of insulating polystyrene covers an area of 33 sqm, so the amount of insulating polystyrene used equals 106,970 cubic meters.

Competitors Analysis

After analyzing the market, it was found that most of the used insulation is imported from Saudi Arabia, which explains that the insulation produced locally does not comply with international specifications (pressure 32-35). The following are local factories for insulation materials:

- Insuestrong for insulating materials industry.
- Designs for Engineering Industries.
- Jordan Mountains for Intermediate Industries.
- Al-Wattanyiah for Insulating Materials Industry.
- Jordan River for Polystyrene products.
- Al-Alamyiah Al-awla for Insulating.
-

Production Capacity

The planned annual production capacity of the project based on the plant production line is 43200 cubic meters.

The Projected Production Proportion

The following table shows the percentage and quantity of the projected production for the proposed project:

Table 6 : Projected Production Proportion

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production Proportion	35%	40%	45%	50%	50%	55%	60%	65%	70%	75%
Quantity (m³)	15,120	17,280	19,440	21,600	21,600	23,760	25,920	28,080	30,240	32,400

Price Analysis and Pricing Policy

After studying the local market, it was found that the price of a cubic meter of the extruded polystyrene is 60 JD.

Source: Average prices of the local and imported polystyrene.

The following table shows the cost of the required raw materials in one month:

Table7 : The Required Raw Material

Raw Materials	Percentage	Quantity (KG)	Cost (JD / KG)
Coloring material	%1	512.16	2.5
Gas	%10	7,300.88	1.3
Raw polystyrene	%85	43,533.6	0.91
Powder	%0.5	256.08	0.5
Flame Resistant	%2.5	1280.0	3.0
Others	1%	512.16	1.0
Packing	1 KG/m ³	2340	1.50

Projected Revenues Analysis

- The expected revenues for 10 years are as follows:

Table 8 : Projected Revenues

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production Proportion	35 %	40%	45%	50%	50%	55%	60%	65%	70%	75%
Annual Revenues	907200	1036800	1166400	1296000	1296000	1425600	1555200	1684800	1814400	1944000

4. Technical Analysis

The Required Human Resources

Table 9 : The Required Manpower

Job Title	No.
Indirect Employees	
Plant Manager	1
Financial Manager	1
Public Relations Officer	1
Marketing Employee	1
Guard	1
Cleaner	2
Total Number of Indirect Employees	7
Direct Employees	
Production Line Manager	1
Quality Checker	1
Production Workers	8
Maintenance Workers	2
Driver	2
Total Number of Direct Employees	14
Total Number of Employees	21

Job Description:

- **General Manager:** Will propose the public policy of the Plant and the preparation of plans and follow up on their implementation, technical and administrative supervision of staff, and development of procedures and ensuring occupational health and safety requirements.
- **Financial Manager:** Will be responsible for all financial transactions in the plant.
- **Public Relations Officer :** Prepare work plans and programs, organize and coordinate the procedural processes, work to strengthen the relationship between the factory and the categories it deals with through various media and awareness means, prepare technical reports, manage subordinates and develop their skills, and develop procedures and secure occupational safety and health requirements.
- **Marketing Officer:** carry out the required marketing work according to the approved plan and achieve the best possible spread of the products of the factory and commensurate with its status.
- **Guard:** Implement the approved security instructions to maintain the factory security system.
- **Production Line Officer:** Plan and develop the production plan, prepare the necessary specifications for production, monitor and control the production lines and supervising the production departments, and coordinate the work between them.
- **Maintenance technician:** Read the drawings and installation instructions for machines and switches, operating systems and maintenance instructions, carry out preventive maintenance programs, installation and operation of electrical machines and switches according to the drawings, and inspection of electrical machines.
- **Quality Officer:** Check the quality of the product and conform to the required specifications and standards.
- **Driver:** Driving vehicles to carry out the assigned delivery tasks to him.
- **Cleaner:** Keep the factory clean.

Site Analysis

The proposed site for this project is Al Hasan Industrial City in Irbid, which is the industrial investment center in the North.

Production Process Description

Description of the production of blue polystyrene:

- The pressure of the blue polystyrene ranges between (32-35), depending on the temperature and time it is exposed to.
- Raw material is imported from foreign markets (Turkey, Saudi Arabia...)
- Granules are adsorbed through a machine that uses water vapor at 200 ° C.
- These granules are cooled after doubling their size through cooling tanks.
- These granules are placed in molds to form blocks of different sizes.
- These blocks are cut to different measurements according to customer request through thermal scissors.



Technical Requirements Analysis

Land and construction works

The project will be constructed on land with an area of 4,000 sqm. The following table shows the required land:

Table10 : Required Land

Land	Area (sqm)
The Required land	4,000

The required construction works for the project is represented as follows:

Table 11: Required Construction Works

Construction Works	Area (sqm)
Hangar	1,500
Designing and drawing	1,500
Paving the ground	1,500
Buildings	250

Machinery and Equipment

The following table provides the machinery and equipment needed for the project:

Table 12 : Required Machinery and Equipment

Machinery and Equipment	Quantity
Fire-Fighting Equipment	1
Polystyrene Production Line	1
Cables	1
Electrical Transformer	1
Water Pumps	2
Air Pumps	2

Machinery and Equipment	Quantity
Laboratory Tools	1
Water Cooling System	1
Furnace	1
Computers	10
Forklift	2

Furniture

The project needs furniture for administrative offices.

Vehicles and Transportation

The project needs Vehicles as following:

Table 13: Vehicles and Transportation

Vehicles	Quantity
Pickup Trucks	2

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.81% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 14: General Assumption

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 15: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 16: Annual Growth Rates Assumptions

	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 17: Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	42%
Utilities Expenses from Total Revenues	4.0%
Maintenance Expenses from Total Revenues	1%
Depreciation Expenses from Total Revenues	1.0%
Insurance Expenses from Total Assets Value	1.00%
Marketing Expenses from Total Revenues	1%

Table 18: Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	0%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%

Table 19 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%5
Bets	1.00
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	1,086,148
Loans Percentage	%50
Equity	
Cost of Equity	%16.6
Equity Value	1,086,148
Equity Percentage	%50
Gross	
Project Value	2,172,295
Weighted Average Cost of Capital	%100

Capital Expenditures

The estimated capital costs of the project are 2,172,295 JD and cover the cost of land, construction, machinery and equipment, furniture and furnishings, vehicles and transportation, pre-operating expenses and initial working capital. These costs are shown in the following tables:

Table 20: Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost (JD)
Required Land	15	4,000	60,000
Gross Cost			60,000

Table 21: Construction Work Capital Expenditure

Construction Works	Cost (JOD per Squared	Area	Gross Cost / Jordan
Hanger	100		150,000
Designing and planning	7		10,500
Paving	25		37,500
Construction Works	250		62,500
Gross Total			260,500

Table 22: Machinery and equipment Capital Expenditure

Machinery and equipment	Quantity	Unit cost	Gross Cost/ Jordan Dinar
Fire-Fighting Equipment	1	500	500
Polystyrene Production Line	1	1,250,000	1,250,000
Cables	1	20,000	20,000
Electrical Transformer	1	20,000	20,000
Water Pumps	2	5,000	10,000
Air Pumps	2	5,000	10,000
Laboratory Tools	1	30,000	30,000
Water Cooling System	1	35,000	35,000
Furnace	1	15,000	15,000
Computers	10	300	3,000
Forklift	2	12,000	24,000
Gross Total			1,417,500

Table 23: Furnitutre Capital Expenditure

Furnitutre	Quantity	Unit cost	Gross Cost/ Jordan Dinar
Office Furnitutre	1	7,000	7,000
Total			7,000

Table 24: Vehicles and Transportation Capital Expenditure

Vehicle and Transportation	Quantity	Unit cost	Gross Cost/ Jordan Dinar
Pickup Truck	2	23,000	46,000
Gross Total			46,000

Table 25: Pre-operating expenses

Pre-operating expenses	Gross Cost (JD)
Governmental fees	1,500
Transportation	1,000
Marketing	4,000
Training	2,000
Total	8,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 26: Initial working capital

Initial working capital	Number of months	Percentage	Cost (JD)
Operating expenses	3	%25	144,390
Administrative and general expenses	3	%25	6,655
Marketing expenses	3	%25	2,268
Indirect salaries	3	%25	22,000
Gross Total			175,314

Table27 : Summary of expenses

Summary of capital expenditures	Cost (JD)	Contingency	Cost (JD)	Percentage
Land	60,000	10%	66,000	%3.0
Construction Works	260,500	10%	286,550	%13.2
Machinery and Equipment	1,417,500	10%	1,559,250	%71.8
Furniture	7,000	10%	7,700	%0.4
Vehicles and Transportation	46,000	10%	50,600	%2.3
Pre-operating expenses	8,500	10%	9,350	%0.4
Initial working capital	175,314	10%	192,845	%8.9
Total (JD)	1,974,814		2,172,295	100%

Sources of funding

The financing structure required for the project consists of loans and equity. The investment cost of this project was estimated at 2,172,295 JD. This amount is determined through fixed assets, which include land, construction, machinery and equipment, furniture and furnishings, vehicles and transport vehicles, pre-operating expenses and initial working capital. (50%) of the project will be financed through loans (1,086,148 JD) and the rest (50%) through self financing (equity). The following table shows the general structure of the required financing:

Table 28: Sources and use of finance

Sources of funding	amount	Percentage
Equity (self-financing)	1,086,148	%50
Loans	1,086,148	%50
Gross Total	2,172,295	%100
Use of funding	amount	Percentage
Capital expenditure	1,970,100	%90.69
Pre-operating expenses	9,350	%0.43
Working capital	192,845	%8.88
Gross Total	2,172,295	%100.00

Operating Expenses

The following table shows the expected operating expenses of the project:

Table 29: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw material expenses	381,024	435,456	489,888	544,320	544,320	598,752	653,184	707,616	762,048	816,480
Service expenses	36,288	41,472	46,656	51,840	51,840	57,024	62,208	67,392	72,576	77,760
Maintenance expenses	9,072	10,368	11,664	12,960	12,960	14,256	15,552	16,848	18,144	19,440
Consumables	9,072	10,368	11,664	12,960	12,960	14,256	15,552	16,848	18,144	19,440
Direct salaries	89,600	93,274	97,098	101,079	105,223	109,537	114,028	118,703	123,570	128,637
Other expenses	52,506	59,094	65,697	72,316	72,730	79,383	86,052	92,741	99,448	106,176
Gross Total	577,562	650,031	722,667	795,475	800,033	873,208	946,577	1,020,148	1,093,930	1,167,932

General and Administrative Expenses

The following table shows the expected administrative and general expenses of the project:

Table30 : administrative and general expenses

description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom expenses	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality expenses	500	515	530	546	563	580	597	615	633	652
Stationery and prints	500	515	530	546	563	580	597	615	633	652
Transportation and movements	1,800	1,854	1,910	1,967	2,026	2,087	2,149	2,214	2,280	2,349
Rent expenses	0	0	0	0	0	0	0	0	0	0
Training expenses	500	515	530	546	563	580	597	615	633	652
Insurance expenses	19,701	20,292	20,901	21,528	22,174	22,839	23,524	24,230	24,957	25,705
Other expenses	2,420	2,493	2,567	2,645	2,724	2,806	2,890	2,976	3,066	3,158
Gross Total	26,621	27,420	28,242	29,090	29,962	30,861	31,787	32,741	33,723	34,734

Marketing Expenses

Table 31: Marketing Expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	9,072	10,368	11,664	12,960	12,960	14,256	15,552	16,848	18,144	19,440

Human Resources

Table 32: Expected staff preparation

Job title	No.	Monthly salaries	Total monthly	Total yearly salaries
Indirect Employees				
General Manager	1	2,000	2,000	32,000
Financial Manager	1	1,500	1,500	24,000
Public Relations Officer	1	750	750	12,000
Marketing Employee	1	450	450	7,200
Guard	1	300	300	4,800
Cleaner	2	250	500	8,000
Total Indirect Employees	7		5,500	88,000
Direct Employees				
Production Line Officer	1	900	900	14,400
Quality Assurance	1	700	700	11,200
Production Employee	8	300	2,400	38,400
Maintenance Technicians	2	400	800	12,800
Driver	2	400	800	12,800
Total Direct Employees	14		5,600	89,600
Total Employees	21		11,100	177,600

The salaries and monthly wages of each employee were estimated to be multiplied by 16 months of the year for the purpose of calculating the increase in salaries and annual wages due to social security, health insurance, etc. to the employees of the administration. The following table shows the expected annual salaries and wages. The annual rate of increase in wages and salaries is taken into account:

Table 33: Expected monthly salaries for the years (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	2,000	2,082	2,167	2,256	2,349	2,445	2,545	2,650	2,758	2,871
Financial Manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Public Relations Officer	750	781	813	846	881	917	954	994	1,034	1,077
Marketing Employee	450	468	488	508	528	550	573	596	621	646
Guard	300	312	325	338	352	367	382	397	414	431
Cleaner	250	260	271	282	294	306	318	331	345	359
Direct Employees										
Production Line Officer	900	937	975	1,015	1,057	1,100	1,145	1,192	1,241	1,292
Quality Assurance	700	729	759	790	822	856	891	927	965	1,005
Production Employee	300	312	325	338	352	367	382	397	414	431
Maintenance Technicians	400	416	433	451	470	489	509	530	552	574
Driver	400	416	433	451	470	489	509	530	552	574

Table 34: Expected Yearly salaries for the years (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
Financial Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Public Relations Officer	12,000	12,492	13,004	13,537	14,092	14,670	15,272	15,898	16,550	17,228
Marketing Employee	7,200	7,495	7,803	8,122	8,455	8,802	9,163	9,539	9,930	10,337
Guard	4,800	4,997	5,202	5,415	5,637	5,868	6,109	6,359	6,620	6,891
Cleaner	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Total Indirect Salaries	88,000	91,608	95,364	99,274	103,344	107,581	111,992	116,584	121,364	126,340
Direct Employees										
Production Line Officer	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
Quality Assurance	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Production Employee	38,400	39,974	41,613	43,319	45,096	46,945	48,869	50,873	52,959	55,130
Maintenance Technicians	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Driver	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Total Direct Salaries	89,600	93,274	97,098	101,079	105,223	109,537	114,028	118,703	123,570	128,637
Gross Total	177,600	184,882	192,462	200,353	208,567	217,118	226,020	235,287	244,934	254,976

Depreciations

The following tables show all equipment costs, expenses and depreciation rates

Table 35: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	66,000	%0.0	%0.0
Construction Works	286,550	%5.0	%0.0
Machinaries	1,559,250	%10.0	%2.0
Furniture	7,700	%10.0	%5.0
Vehicles	50,600	%10.0	%0.0
	1,970,100		

Table 36: Depreciation and annual increases on construction work

Summary of capital costs, additions and annual depreciation	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total fixed assets	1,970,100	2,001,670	2,033,883	2,066,752	2,100,292	2,134,515	2,169,437	2,205,073	2,241,436	2,278,543
Total consumption expenses	176,083	179,240	182,461	185,748	189,102	192,524	196,016	199,580	203,216	206,927
Total accumulated depreciation	176,083	355,322	537,783	723,531	912,632	1,105,156	1,301,172	1,500,752	1,703,968	1,910,895
Total additions	0	31,570	32,213	32,869	33,539	34,224	34,922	35,635	36,363	37,107
Total net book value	1,794,018	1,646,348	1,496,100	1,343,222	1,187,660	1,029,359	868,265	704,321	537,468	367,648

Loans

The following table represents the annual loan amount and the annual payment method with the details of the remaining amounts of loans for each year.

Table 37: Loan Details

Loan Value	1,086,148
Annual Interest Rate	9.48%
Loans Period	5
Grace Period	0
Loan Starts at year:	2018

Year	Annual Payment	Interest	Equity capital	Loan Remaining Value
2018				1,086,148
2019	282,727	102,967	179,760	906,388
2020	282,727	85,926	196,801	709,586
2021	282,727	67,269	215,458	494,128
2022	282,727	46,843	235,883	258,245
2023	282,727	24,482	258,245	0

Income Statement

Table 38: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	907,200	1,036,800	1,166,400	1,296,000	1,296,000	1,425,600	1,555,200	1,684,800	1,814,400	1,944,000
Gross Operating Revenues	907,200	1,036,800	1,166,400	1,296,000	1,296,000	1,425,600	1,555,200	1,684,800	1,814,400	1,944,000
Operating Expenses	-577,562	-650,031	-722,667	-795,475	-800,033	-873,208	-946,577	-1,020,148	-1,093,930	-1,167,932
Gross Operating Profit	329,638	386,769	443,733	500,525	495,967	552,392	608,623	664,652	720,470	776,068
Gross Profit Percentage	%36	%37	%38	%39	%38	%39	%39	%39	%40	%40
Salaries and Benefits (Indirect Staff)	-88,000	-91,608	-95,364	-99,274	-103,344	-107,581	-111,992	-116,584	-121,364	-126,340
General and Administraive Expenses	-26,621	-27,420	-28,242	-29,090	-29,962	-30,861	-31,787	-32,741	-33,723	-34,734
Markeing Expenses	-9,072	-10,368	-11,664	-12,960	-12,960	-14,256	-15,552	-16,848	-18,144	-19,440
Pre Operating Expenses	-9,350									
Gross Indirect Expenses	-133,043	-129,396	-135,270	-141,323	-146,266	-152,698	-159,331	-166,172	-173,230	-180,514
Income before Interest, Depreciation and Tax	196,595	257,373	308,463	359,202	349,700	399,694	449,292	498,480	547,239	595,554
Fixed Assets Depreciations	-176,083	-179,240	-182,461	-185,748	-189,102	-192,524	-196,016	-199,580	-203,216	-206,927
Income before Tax and Interests	20,513	78,133	126,002	173,454	160,599	207,170	253,276	298,900	344,023	388,627
Bank Interests	-102,967	-85,926	-67,269	-46,843	-24,482	-0	-0	-0	-0	-0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Before Tax	-82,454	-7,792	58,733	126,611	136,117	207,170	253,276	298,900	344,023	388,627
Income Tax	0	0	-2,721	-6,331	-6,806	-10,358	-12,664	-14,945	-17,201	-19,431
Net Profit	-82,454	-7,792	56,013	120,280	129,311	196,811	240,612	283,955	326,822	369,196
Net Profit Percentage	%9-	%1-	%5	%9	%10	%14	%15	%17	%18	%19
Compulsory Reserves	0	0	-5,601	-12,028	-12,931	-19,681	-24,061	-28,395	-32,682	-36,920
Retained Earnings	-82,454	-90,246	-39,835	68,418	184,798	361,928	578,479	834,038	1,128,178	1,460,454

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 39: Expected Cashflows Statement

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	-82,454	-7,792	56,013	120,280	129,311	196,811	240,612	283,955	326,822	369,196
Non-monetary adjustments										
Bank Interests	102,967	85,926	67,269	46,843	24,482	0	0	0	0	0
Depreciation	176,083	179,240	182,461	185,748	189,102	192,524	196,016	199,580	203,216	206,927
Total Operating Cashflows before Additions to Working Capital	196,595	257,373	305,742	352,871	342,894	389,335	436,629	483,535	530,038	576,122
Inventory (Increase/Decrease)	-31,752	-4,536	-4,536	-4,536	0	-4,536	-4,536	-4,536	-4,536	-4,536
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	32,756	3,006	3,034	3,063	760	3,124	3,156	3,190	3,225	3,262
Working Capital (Increase/Decrease)	1,004	-1,530	-1,502	-1,473	760	-1,412	-1,380	-1,346	-1,311	-1,274
Net Cashflows from Operating Activities	197,600	255,843	304,240	351,398	343,654	387,923	435,249	482,189	528,727	574,848
Cashflows from Investments Activities										
Fixed Assets (Procurement)	-1,970,100	-31,570	-32,213	-32,869	-33,539	-34,224	-34,922	-35,635	-36,363	-37,107
Net Cashflows from Investments Activities	-1,970,100	-31,570	-32,213	-32,869	-33,539	-34,224	-34,922	-35,635	-36,363	-37,107

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cashflows from Financing Activities										
Capital	1,086,148									
Loan Amortization	-179,760	-196,801	-215,458	-235,883	-258,245	0	0	0	0	0
Bank Interest Rate	-102,967	-85,926	-67,269	-46,843	-24,482	-0	-0	-0	-0	-0
Loans	1,086,148	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,889,568	-282,727	-282,727	-282,727	-282,727	0	0	0	0	0
Net increase (decrease) in cash	117,068	-58,454	-10,699	35,802	27,388	353,700	400,327	446,553	492,364	537,741
Cashflows at the Beginning of Period	0	117,068	58,614	47,915	83,717	111,105	464,804	865,131	1,311,684	1,804,048
Cashflows at the end of Period	117,068	58,614	47,915	83,717	111,105	464,804	865,131	1,311,684	1,804,048	2,341,789

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 40: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	117,068	58,614	47,915	83,717	111,105	464,804	865,131	1,311,684	1,804,048	2,341,789
Inventory	31,752	36,288	40,824	45,360	45,360	49,896	54,432	58,968	63,504	68,040
Total assets traded	0	0	0	0	0	0	0	0	0	0
Non-current assets	148,820	94,902	88,739	129,077	156,465	514,700	919,563	1,370,652	1,867,552	2,409,829
Fixed assets (net)										
Total non-current assets	1,794,018	1,646,348	1,496,100	1,343,222	1,187,660	1,029,359	868,265	704,321	537,468	367,648
Total assets	1,794,018	1,646,348	1,496,100	1,343,222	1,187,660	1,029,359	868,265	704,321	537,468	367,648
Liabilities	1,942,837	1,741,250	1,584,839	1,472,299	1,344,124	1,544,059	1,787,828	2,074,973	2,405,020	2,777,477
Current liabilities										
Payables	32,756	35,763	38,796	41,859	42,619	45,743	48,899	52,089	55,314	58,575
Remaining amount of Loan	196,801	215,458	235,883	258,245	0-	0-	0-	0-	0-	0
Total current liabilities	229,557	251,221	274,680	300,104	42,619	45,743	48,899	52,089	55,314	58,575

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-current liabilities										
long term loans	709,586	494,128	258,245	0	0	0	0	0	0	0
Total long - term liabilities	709,586	494,128	258,245	0	0	0	0	0	0	0
Total liabilities	939,144	745,349	532,925	300,104	42,619	45,743	48,899	52,089	55,314	58,575
Property rights										
Members Contributions	1,086,148	1,086,148	1,086,148	1,086,148	1,086,148	1,086,148	1,086,148	1,086,148	1,086,148	1,086,148
Compulsory reserve	0	0	5,601	17,629	30,560	50,242	74,303	102,698	135,380	172,300
Profit (loss)	82,454-	90,246-	39,835-	68,418	184,798	361,928	578,479	834,038	1,128,178	1,460,454
Total equity	1,003,694	995,901	1,051,914	1,172,194	1,301,505	1,498,317	1,738,929	2,022,884	2,349,706	2,718,901
Total liabilities and equity	1,942,837	1,741,250	1,584,839	1,472,299	1,344,124	1,544,059	1,787,828	2,074,973	2,405,020	2,777,477

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 41: Free Net Cash flows Table

Net Cashflows	ما قبل التشغيل	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	القيمة النهائية
Net Free Cashflows	-2,172,295	206,950	224,273	272,027	318,529	310,115	353,700	400,327	446,553	492,364	537,741	6,345,746
Discount Factor	1.00	0.89	0.79	0.70	0.62	0.55	0.49	0.43	0.38	0.34	0.30	0.30
Net Present Value for Cash Flows	-2,172,295	183,445	176,222	189,468	196,658	169,718	171,585	172,147	170,216	166,362	161,058	1,900,602

Table below (38) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 42: payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-2,172,295	206,950	224,273	272,027	318,529	310,115	353,700	400,327	446,553	492,364	537,741
Project Value Returned	2,172,295	1,965,345	1,741,072	1,469,045	1,150,516	840,402	486,702	86,376	-360,178	-852,541	-1,390,282
Payback Period (Year)	7	1	1	1	1	1	1	1	0	0	0

Table 43: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.81
Net Present Value for Cashflows	1,485,185
Payback Period	7
Internal Rate of Return	%19.71

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 44: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%19.71	7	12.81%
Revenues declined by 10%	%16.87	8	12.81%
Operating Expenses Increased by 10%	%15.74	8	12.05%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 1,485,185 Jordan Dinars considering that the project provides 26 Job Opportunities for the governorate residents.